Fiscal Year:	FY 2024	**Task Last Updated:**	FY 04/04/2024
PI Name:	Gibble, Kurt Ph.D.		
Project Title:	Contributions to the Evaluation of the ACES Clock PHARAO		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	16802-6300	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2011 Fundamental Physics NNH11ZTT001N
Start Date:	06/08/2012	**End Date:**	09/30/2025
No. of Post Docs:		**No. of PhD Degrees:**	2
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JPL
Contact Monitor:	Callas, John	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: New contract number is JPL 1631717 per J. Callas/JPL (Ed., 8/29/24) NOTE: End date changed to 09/30/2025 per J. Callas/JPL (Ed., 4/4/24) NOTE: End date changed to 5/31/2023 per U. Israelsson/JPL (Ed., 1/28/2020) NOTE: End date changed to 12/31/2021 per U. Israelsson/JPL (Ed., 1/8/19) NOTE: End date changed to 12/31/2018 per PI (Ed., 3/10/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	JPL 1631717 ; JPL 1459238		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	ACES (Atomic Clock Ensemble in Space) is a European Space Agency (ESA) project that will test fundamental physics, search for the time variation of fundamental constants, and enable highly precise frequency comparisons between clocks on Earth. It is planned to operate for at least 18 months on the International Space Station. PHARAO (Projet d'Horloge Atomique par Refroidissement d'Atomes en Orbit) is the laser-cooled, microgravity, atomic cesium clock of ACES that provides state-of-the-art accuracy, with a goal of 1 part in 10^(16) accuracy. We will assist in the accuracy evaluation of PHARAO, a central scientific objective of the ESA mission. We will contribute to the analysis of the three largest sources of systematic uncertainty: clock frequency shifts due to the collisions of ultracold atoms, a Doppler shift known as the distributed cavity phase shift, and a frequency shift due to the lensing of the atomic wavepackets by the clock's microwave field. The initial working uncertainty budget projected an accuracy of 2.2×10^(-16). The proposed work will reduce the largest contribution to the clock's uncertainty to negligible levels and investigate significant reductions of others, with a goal of reaching 1×10^(-16) accuracy. The work will help define the limits of PHARAO's accuracy and identify workable procedures to demonstrate its accuracy. The PI will also participate in a study on space optical clocks as a member of NASA's science definition team (SDT) for optical clocks and time transfer.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our work on microwave cavities produced an improved design that will be used for Cs and Rb fountain clocks. These are used in new clocks at the British, Canadian, South Korean, Polish, and Indian National Laboratories, and at CERN. The new cavity design allows significantly reduced clock errors due to Doppler shifts. Our work has also led to a deeper understanding of the microwave lensing frequency shift that is now being used as a correction of International Atomic Time, including the correction for the PHARAO clock's frequency. More broadly, our work contributes to scientific progress towards anticipated future optical frequency atomic clocks, including testing fundamental physics with state-of-the-art optical clocks in space.
Task Progress:	The launch date of the ACES mission is currently set for February 2025. In late 2020, a pre-launch accuracy evaluation of the PHARAO clock was published, which completed the bulk of the Pennsylvania State University (Penn State) work expected before launch. The Penn State group continues their contact with the PHARAO and ACES teams to address any issues that may arise, and maintains their software for PHARAO, for the reaming work to establish the PHARAO clock accuracy once in orbit. Additional work this year for PHARAO examined the uncertainty of the combined velocity and density extrapolation and the impact of correlations on the instability of the extrapolated PHARAO frequency, the instability of the extrapolated PHARAO frequency correction for its high stability operating point, and the instability of the systematic uncertainty of the corrected PHARAO frequency (at its high stability operating point). An invited talk on this work was presented at the ACES Workshop 2023, hosted by the German Space Agency (DLR) in Ulm, Germany. As a member of the NASA science definition team (SDT) for optical clocks and time transfer, Principal Investigator (PI) Gibble presented two invited talks and an invited poster on the "Fundamental physics with an Optical Clock Orbiting in Space" (FOCOS) mission concept at the 2023 NASA Fundamental Physics Workshop, the 9th Symposium on Frequency Standards and Metrology, and the 2024 FunPAG Virtual Science Workshop. A colloquium at the Universität Bonn discussed PHARAO/ACES, FOCOS, and the broader work of Gibble’s group.
Bibliography Type:	Description: (Last Updated: 05/28/2026)
Abstracts for Journals and Proceedings	Gibble K. "FOCOS mission concept." 2023 NASA Fundamental Physics Workshop, Santa Barbara, California, May 23-25, 2023. Abstracts. 2023 NASA Fundamental Physics Workshop, Santa Barbara, California, May 23-25, 2023. , May-2023
Abstracts for Journals and Proceedings	Gibble K. "Precision Measurements and Fundamental Physics II." 9th Symposium on Frequency Standards and Metrology, Kingscliff, NSW, Australia, October 16-20, 2023. Abstracts. 9th Symposium on Frequency Standards and Metrology, Kingscliff, NSW, Australia, October 16-20, 2023. , Oct-2023